Project Progress Report: Canteen Management System

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1. Project Overview and Current Status

1.1 Project Summary

The Canteen Management System is a web-based application designed to streamline food ordering and management within a canteen environment. The system allows users to browse menu items, place orders, and track their requests. Administrators can efficiently manage menu items and monitor customer orders through a centralized dashboard.

Technologies Used

- Laravel (PHP Framework)
- MySQL Database
- Blade Template Engine
- Bootstrap CSS

Current Completion Status

Approximately 70% completed

High-Level Status Statement

The core backend functionalities, including user authentication, menu management, and order processing, have been successfully implemented. Frontend integration is ongoing, with most major components already functional. Remaining work focuses on UI refinement and advanced features such as order tracking.

1.2 Progress Against Schedule

Status: Slightly Delayed

Reason for Deviation:
The project faced minor delays due to challenges in managing complex database relationships and implementing secure authentication. These issues have now been resolved, and development is progressing steadily.

2. Detailed Development Progress

2.1 Backend and Database Implementation

The backend is developed using Laravel following the MVC (Model-View-Controller) architecture to ensure scalability and maintainability.

Database Schema

The following core tables have been implemented:

- Users – Stores user credentials and roles
- Items – Contains all food menu items
- Orders – Stores order details
- Order_Items – Manages many-to-many relationships between orders and items

Authentication Module

- User registration, login, and logout implemented
- Password hashing and session management ensured

Status: Complete

Data CRUD APIs

- CRUD operations for menu items and orders implemented

Status: 80% Complete

API Endpoints

- POST /register
- POST /login
- GET /menu
- POST /order
- GET /orders

Key Feature Implemented

A dynamic order management system allowing users to:

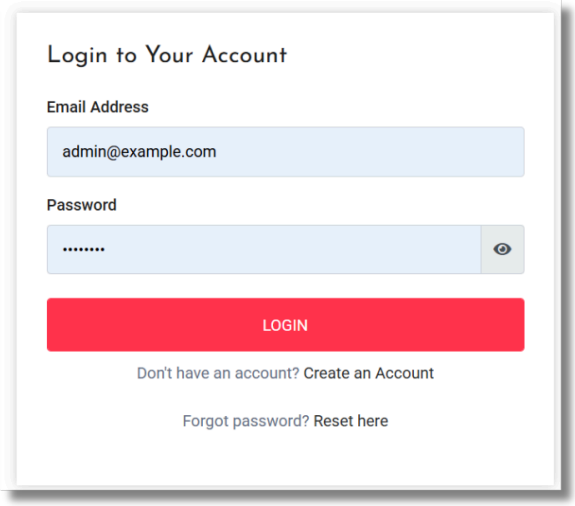
- Select multiple food items
- Place orders
- View orders in both user dashboard and admin panel

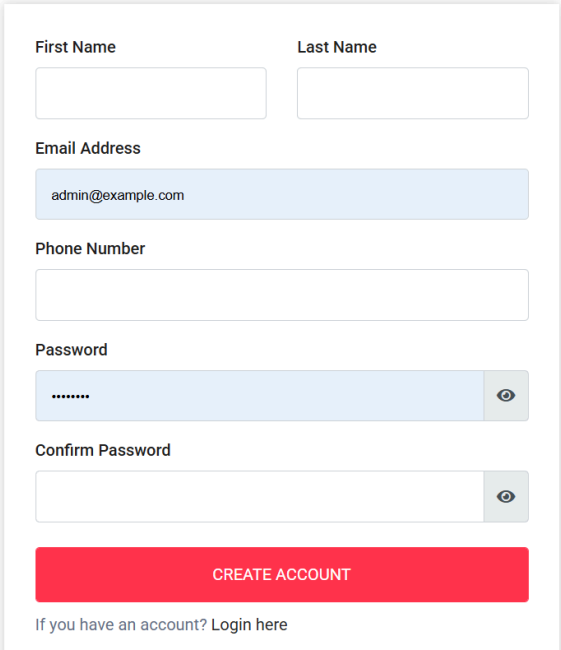
2.2 Frontend Implementation

The frontend is built using Blade templates and Bootstrap, ensuring responsiveness and usability.

Login & Registration Pages

- Fully functional and integrated with backend

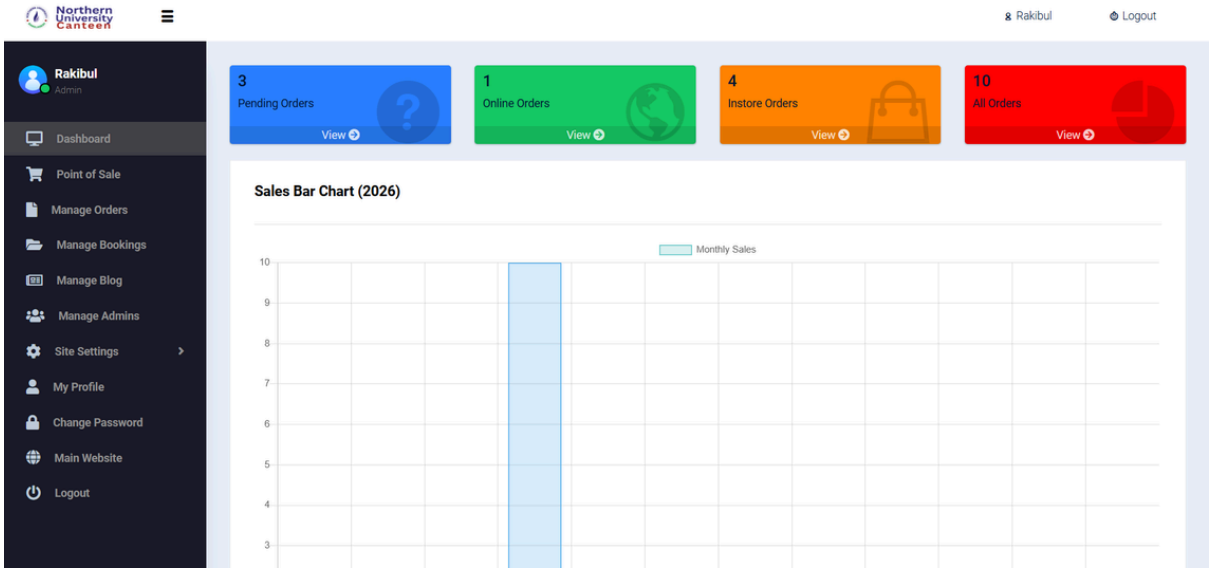




Integration Status: Yes

Main Dashboard / Home Page

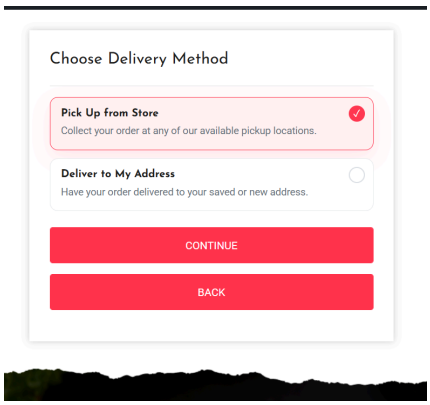
- Displays menu items
- Enables user interaction

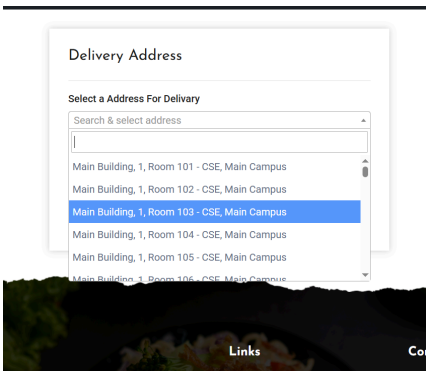


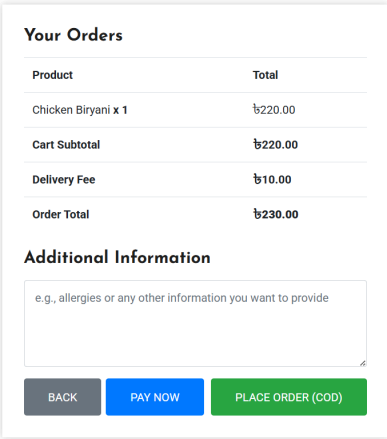
Integration Status: Yes

Order Form

- Interactive form for selecting and ordering items







Integration Status: Yes

Navigation & Layout

- Responsive navbar
- Consistent UI layout using Bootstrap

Status: Complete

UI Status

The UI structure is complete. Further improvements are being made to enhance:

- User experience (UX)
- Visual design (UI)

3. Verification and Testing

3.1 Testing Summary

Unit Testing

- Tested authentication and order modules

Integration Testing

- Verified communication between frontend, backend, and database

Example Test Case

Scenario: Successful Order Placement

Steps:

1. User logs into the system
2. Selects food items
3. Submits the order
4. Data is stored in the database

Result:
Order is successfully recorded and visible in both user dashboard and admin panel

3.2 Identified Issues / Bugs

Issue	Description	Status
Duplicate Order Submission	**Caused by page refresh**	**Resolved**
Responsive UI Issues	**Layout issues on small screens**	**Pending**

4. Technical Challenges and Solutions

4.1 Challenge 1: Managing Database Relationships

Description:
Handling relationships between orders and multiple items was complex.

Solution:
Implemented Laravel Eloquent ORM relationships:

- One-to-Many
- Many-to-Many

4.2 Challenge 2: Secure Authentication

Description:
Ensuring a secure login system was initially challenging.

Solution:
Used Laravel's built-in authentication system:

- Secure password hashing
- Session management

5. Work Plan and Next Steps

5.1 Remaining Tasks

- Integrate payment system (optional/simulated)
- Improve UI/UX design
- Develop admin analytics dashboard
- Final testing and bug fixing

5.2 Proposed Timeline

Task	Estimated Completion Date
Core Feature Completion	**1-5-2025**
Testing & Bug Fixing	**2-5-2026**
Final Documentation & Deployment	**5-52026**